Article

UzABSA-LLM: Parameter-Efficient Fine-Tuning and Multi-Domain Annotation for Uzbek Aspect-Based Sentiment Analysis

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Abstract

In this paper, we investigate the application of parameter-efficient fine-tuning to scalable aspect-based sentiment analysis (ABSA) of Uzbek business reviews. More concretely, we fine-tune Qwen 2.5-7B, Llama 3.1-8B, and DeepSeek-R1-Distill-7B in one QLoRA setup to perform aspect term extraction and polarities classification in pairs, giving results in the format of structured JSON. Our evaluation is based on a test set of 609 samples taken from the review-grouped and deduplicated split, using zero-shot prompting with the same models as the control condition. As a result of fine-tuning, we obtain nearly double F1 for aspect term extraction and triple F1 for aspect-polarity pair classification: Qwen has the best extraction F1 (0.708) with a 100% parsing ratio, whereas Llama leads in pair F1 (0.651) and sentiment classification accuracy (0.925). We then compare the performance of our fine-tuned LLMs to that of the fine-tuned 110M-parameter Uzbek BERT encoder on the task of sentiment classification, joint prediction, and aspect extraction. LLMs surpass BERT models in sentiment classification and joint prediction tasks but are comparable in the aspect extraction task. We outline the three-step annotation process: local LLM-based annotation, LLM-as-Judge-based score generation, and aggregation based on quality tiers. This process results in 9,884 aspect-level annotations when used on 5,038 business review documents across 23 domains. The scores generated by the LLM judge correlate with the human scores at $\rho = 0.66$. Furthermore, we additionally release a reconciled, double-annotated subset of 80 reviews. All code, models, and datasets are publicly released.

Keywords: Aspect-Based Sentiment Analysis; Big Data Text Analytics; Low-Resource NLP; Uzbek Language; Parameter-Efficient Fine-Tuning; QLoRA; Large Language Models; LLM-as-Judge; Multi-Domain Dataset

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1. Introduction

Aspect-Based Sentiment Analysis (ABSA) identifies certain aspects and polarity in text [1]. While high-resource languages dominate the field, Uzbek which is a Turkic language family with nearly 38 million speakers[1], is comparatively under-explored, owing to script duality (Latin/Cyrillic), agglutinative morphology, and limited representation in

[1] https://www.worldometers.info/world-population/uzbekistan-population/

multilingual pre-training corpora. Recent 7 to 8B-parameter LLMs, such as Qwen 2.5 [2], Llama 3.1 [3], and DeepSeek-R1 [4], provide cross-lingual transfer potential. We use QLoRA [5] to adapt these models on consumer hardware, reformulating ABSA as a single-pass JSON generating operation.

This paper addresses three research questions. **RQ1:** Can QLoRA fine-tuning of an open-weight large language model perform joint aspect extraction and sentiment classification for Uzbek? No prior work establishes this, so viability must be settled before any comparison is meaningful. **RQ2:** Across models fine-tuned under one identical QLoRA setting, does validation loss predict downstream ABSA performance? Checkpoints are routinely selected on loss, yet that proxy has not been validated for structured extraction. **RQ3:** Can a model fine-tuned only on restaurant reviews annotate businesses in other domains? The answer determines whether one annotator can serve a heterogeneous corpus or whether each domain requires its own.

Beyond these language-specific challenges, this study addresses a problem of large-scale text analysis: how to turn a continuous stream of user-generated text into a resource of known quality. Review sites produce far more text than human annotators can label, and in low-resource languages trained annotators are scarce to begin with. We propose a three-step procedure. First, a fine-tuned model annotates the entire corpus. Second, an LLM judge scores the quality of each annotation, and we calibrate that judge against native-speaker ratings rather than assume it is reliable. Third, a tiered assembly step attaches an explicit quality label to every instance in the released dataset. The pipeline is model-agnostic and transfers to other low-resource languages and domains facing the same bottleneck.
In addressing these questions, we make the following contributions:

1. Our work is the first investigation of the performance of open-weight LLMs fine-tuned with the QLoRA method for Uzbek ABSA. The comparison is made between three LLMs in terms of their identical parameter size under the same training setup by using a de-duplicated and review-grouped split of the dataset. We compare our models to those fine-tuned Uzbek BERT encoders and also zero-shot baselines in an unbiased manner.
2. A scalable annotation pipeline assisted by model prediction that employs local inference, LLM judge-based score evaluation, and tiered assembly is proposed. In addition to the scalable pipeline, there is a native speaker validation process which includes two-stage annotation-agreement process, LLM Judge calibration, and double-annotated reconciled subset to estimate the validity of the created resource.
3. The proposed Uzbek ABSA dataset covers 5,038 reviews drawn from 23 business domains, with 9,884 aspect–sentiment pairings. The labeling of this dataset is based on five different categories of aspects proposed by UzABSA; however, a categorization of 119 sub-aspects is also provided.

One observation recurs across all experiments: minimizing cross-entropy loss does not track effectiveness on the ABSA task. The model with the lowest validation loss is not the best aspect extractor, a discrepancy we examine in Section 6. To support reproducible research in low-resource NLP, all code, datasets, and fine-tuned models are publicly available on GitHub[2] and HuggingFace[3].

The remainder of this paper is organized as follows. Section 2 reviews related work on ABSA, parameter-efficient fine-tuning for low-resource languages, and LLM-based evaluation. Section 3 describes the two datasets used in this study. Section 4 presents the methodology, including model selection, the QLoRA configuration, and the annota-

[2] https://github.com/SanatbekMatlatipov/UzABSA-LLM
[3] https://huggingface.co/Sanatbek/UzABSA-LLM

tion pipeline. Section 5 reports the experimental setup and results. Section 6 discusses implications and limitations, and Section 7 concludes with directions for future work.

2. Related Work

SemEval-2014 Task 4 [1] established the modern ABSA paradigm by defining Aspect Term Extraction (ATE) [6], Aspect Category Detection (ACD) [7], Aspect Sentiment Classification (ASC) [8], and joint end-to-end ABSA on English restaurant and laptop reviews [9]. Early systems typically combined sequence labeling (e.g., CRF/BiLSTM) for extraction with attention-based sentiment classifiers. With the rise of pre-trained Transformers, BERT-style models quickly dominated ABSA benchmarks [10,11], casting ATE as token classification and ASC as sentence(-pair) classification via fine-tuning, and neural sentiment classification architectures continue to evolve beyond the standard Transformer encoder [12]. The field has since broadened along two axes relevant to this study: richer structured targets combined with parameter-efficient fine-tuning, where updating under 1% of parameters approaches full fine-tuning on aspect–category–opinion–sentiment extraction [13], and non-English settings, where annotation scarcity has motivated unsupervised or distantly supervised ABSA pipelines, e.g., for Arabic traffic-service reviews [14].

Recent research has increasingly reframed ABSA as a generation task rather than a labeling one. This shift matters for Uzbek in particular: a generative model produces aspect terms and their polarities in one pass, without requiring the span-level tag inventory that supervised sequence labeling would need and that Uzbek largely lacks. The paper of Scaria et al.[15] introduces InstructABSA, which applies instruction-tuned large language models to generate structured text, where both aspects and sentiments can be extracted at once. Simmering and Huoviala [16] compare GPT family models and fine-tuned encoders for English ABSA, observing that fine-tuned small models can outperform significantly larger prompted models. State-of-the-art results have been reached with fine-tuned LLaMA family models for all English ABSA subtasks [17], while zero-shot experiments with nine LLMs on multilingual ABSA subtasks revealed that prompted models cannot overcome fine-tuned baselines [18] yet. Thus, a fair comparison between fine-tuning and zero-shot setting is relevant in this study. Finally, recently released reasoning enhanced models for multilingual ABSA trained on datasets of the size much greater than the size of SemEval-2014 dataset were made available for six higher-resource languages. Despite all those advances, generative ABSA models have almost exclusively been developed for English and other high-resourced languages [19], and morphologically rich low-resource languages have not yet received any attention; this paper tries to fill this gap by fine-tuning open-weight instruction-tuned LLMs on Uzbek.

LLM Fine-Tuning for Low-Resource Languages. Large multilingual foundation models have brought NLP to languages other than English, with architectures like mT5, BLOOM, Llama [3,20], and Qwen [2] differing in their low-resource language coverage because of the differences in pre-training data used. For Turkic languages, Qwen 2.5 provides better multilingual coverage compared to Llama 3.1, although none of the two architectures can guarantee Uzbek support. In low-resource scenarios, adaptation to specific tasks is limited by the computational expenses of full-fledged fine-tuning of 7–8B parameter models, where one needs tens of gigabytes of VRAM and computational resources. The parameter-efficient adaptation techniques allow avoiding such costs—LoRA [21] adds trainable low-rank adapters to the attention heads while keeping all the base weights frozen, and QLoRA [5] uses LoRA along with 4-bit NormalFloat quantization to decrease memory requirements to below 6 GB for 7B-scale models and thus make adaptation possible with only one GPU. Existing research demonstrates excellent results of parameter-efficient adaptation for several non-English or low-resource tasks (such as sentiment analysis of Arabic language texts, named entity recognition in Turkish, or question answering in Korean). The recent works combining LoRA-tuned 7B models with encoder-only baselines for structured information extraction show competitiveness of generation-based models with BERT-style taggers [22] based on supervised learning. Moreover, the studies of domain specialization using different adaptation strategies (fine-tuning vs. retrieval-augmented) over the same open-weight backbone models (Llama 3.1–8B, Qwen 2.5–7B) prove that adaptation strategy is more important than the model itself for performance in downstream [23] tasks. Robustness challenges caused by the transfer of models between different data distributions have been studied in low-resource offensive-language detections [24]. However, this issue has been addressed in our multi-domain evaluation framework. LoRA and QLoRA adaptation of the large language models has not been explored for Uzbek NLP so far.

Uzbek Natural Language Processing. Early work addressed core infrastructure: Matlatipov and Vetulani [25] modeled Uzbek morphology in Prolog, and Uzbek inflectional endings were later packaged as a stand-alone language resource [26]; Aripov et al. [27] studied multilingual MT with Uzbek, highlighting challenges from agglutinative morphology, while Sharipov et al. [28] introduced UzbekTagger, a rule-based POS tagger, and UzMorphAnalyser [29] added morphological analysis over inflectional endings. Script variation is another persistent obstacle: after the 1993 shift from Cyrillic to Latin, much user text still appears in Cyrillic; Salaev et al. [30] provide a machine transliteration tool between the Uzbek alphabets.

Syntactic resources are now extending earlier morphological work on Uzbek. A Universal Dependencies treebank of literary and educational texts provides gold morphosyntactic and dependency annotations [31]. The experiments conducted using the resource make a comparison between graph- and transition-based parsers and BERT encoders [32] and show that cross-treebank learning increases robustness when treebanks of Uzbek language are small-sized [33]. The discovery is important for the field of ABSA, as the aspect terms of Uzbek language are often nominals marked by case and possessive suffixes, and the relation between these aspect terms and opinion expressions can be based on syntax [34]. Our models process raw text without syntactic structure. These resources therefore make syntax-aware Uzbek ABSA a practical direction, which we discuss as future work in Section 7.

For sentiment analysis, Kuriyozov and Matlatipov [35] created an early sentence-level Uzbek dataset with baseline classifiers, later extended by [36] through a broader study of low-resource sentiment dataset construction and benchmarking. Matlatipov et al.[37] further examined sentiment classification of locally collected Uzbek restaurant reviews. The closest resource to our study is UzABSA [38], the first Uzbek ABSA dataset, following the SemEval-2014 Task 4 scheme [1], with 6,175 annotated restaurant sentences, 7,412 aspect-term spans with polarity, and 7,724 aspect-category labels over five categories (*ovqat*, *muhit*, *xizmat*, *narx*, *boshqalar*).

On the modeling side, monolingual Uzbek encoders have recently become available, notably BERTbek [39], trained on Uzbek news text, and TahrirchiBERT [40], trained on $\sim$5B tokens of Latin-script Uzbek; we employ both as fine-tuned baselines in this study (Section 4.2).

Concurrently with this work, Yusufu et al. [34] released LASQ, a low-resource aspect-based sentiment *quadruple* dataset covering Uzbek and Uyghur, paired with a grid-tagging model that injects part-of-speech and dependency knowledge to counter the lexical sparsity of agglutinative languages. LASQ and the present study are complementary rather than competing: LASQ contributes new target–aspect–opinion–sentiment annotations over Google Play reviews and a syntax-aware discriminative tagger, whereas we hold the annotation scheme fixed (the existing SemEval-style UzABSA benchmark) and ask what parameter-efficient adaptation of open-weight generative LLMs delivers on it, relative to monolingual encoders. Their result that syntactic structure helps agglutinative extraction is a natural next ingredient for the models compared here.

Overall, prior Uzbek NLP work, including UzABSA [38], has mainly relied on rule-based methods, traditional machine learning, or encoder-based neural models. To the best of our knowledge, this is the first controlled comparison of QLoRA-adapted open-weight 7–8B models on the UzABSA benchmark, benchmarked against monolingual Uzbek encoders and coupled with a multi-domain annotation case study.

LLM-as-Judge for Annotation Quality. LLM-as-Judge uses large language models as automated evaluators, providing a scalable alternative to human assessment of generated text. Zheng et al.[41] introduced MT-Bench and Chatbot Arena, showing that strong LLMs (e.g., GPT-4) can yield scores that correlate with human preferences in open-ended dialogue. This approach has since been extended to instruction-following, summarisation, and translation, offering cost efficiency, reproducibility, and scalability when human annotation is expensive or impractical—particularly for low-resource languages. However, LLM judges may suffer from positional bias, self-enhancement bias, and session-to-session score drift [42]. While these issues are documented for chat-style evaluation, they are less studied for structured tasks such as information extraction or ABSA.

We use LLM-as-Judge for quality control during dataset construction rather than for ranking systems: the judge decides which machine-generated ABSA annotations are good enough to release, in a low-resource setting where no gold references exist against which to check them. Concretely, GPT-4o-mini scores Uzbek aspect–sentiment outputs on five rubric dimensions (completeness, accuracy, sentiment correctness, relevance, overall quality) using a 1–5 Likert scale. Unlike prior work, our target is structured extraction rather than free-form text, and the judge must understand Uzbek input; we then apply stratified sampling over 23 domains and quality thresholds to guide dataset assembly (Section 4.3). Closest to our pipeline, Hellwig et al. [43] train lightweight ABSA models directly on LLM-generated annotations, establishing model-assisted annotation as a viable route for ABSA; our pipeline differs in adding an explicit judging layer whose scores are themselves calibrated against native-speaker ratings (Section 5.5).

3. Datasets

This study draws on two complementary corpora. The first is the UzABSA dataset [38], a publicly available collection of 6,175 annotated Uzbek restaurant review sentences hosted on HuggingFace.[4] After preprocessing and converting the original 6,175 examples into ChatML-formatted instruction–response pairs, we retain 6,089 usable training instances, which we split 90/10 into 5,480 training and 609 evaluation examples (seed 42); the split groups identical normalised review texts on the same side of the partition, for reasons quantified in Section 5.3. Each sentence is annotated following the SemEval-2014 Task 4 scheme [1]: aspect terms are marked with character-level spans and assigned one of four polarity labels (*positive*, *negative*, *neutral*, *conflict*), and aspect categories are independently labeled with polarity. The dataset contains 7,412 aspect term annotations and 7,724 aspect category annotations distributed across five domain-specific categories: *ovqat* (food), *muhit* (ambiance), *xizmat* (service), *narx* (price), and *boshqalar* (other). On average, each sentence contains 2.53 aspect annotations and 6.42 words.

A notable characteristic of the dataset is class imbalance: positive polarity accounts for 56% of term-level annotations (4,153 of 7,412), followed by neutral (21.6%), negative (21.0%), and conflict (1.4%). This skew toward positive sentiment reflects the natural distribution of

[4] https://huggingface.co/datasets/Sanatbek/aspect-based-sentiment-analysis-uzbek

Table 1. Statistics of the annotated UzABSA dataset used for training and evaluation.

Dataset statistics		Term-level polarity distribution	
Original examples	6,175	Positive	4,153 (56.0%)
Usable ChatML examples	6,089	Neutral	1,601 (21.6%)
Aspect term annotations	7,412	Negative	1,555 (21.0%)
Aspect category annotations	7,724	Conflict	103 (1.4%)
Unique categories	5	Train / Val split	5,480 / 609
Avg. words per sentence	6.42		

restaurant reviews but may bias model predictions toward the majority class. The *conflict* label is dropped during preprocessing (Section 3.2), leaving a three-class polarity space for all experiments. We report macro-averaged sentiment F1 alongside micro-averaged extraction scores to partially account for this imbalance (Section 5). Table 1 summarizes the key statistics.

3.1. Multi-Domain Uzbek Review Corpus for Model-Assisted Annotation

The second corpus is a novel collection of business reviews scraped from Sharh,[5] a publicly accessible Uzbek-language business review platform. After deduplication and minimum-length filtering (removing entries with fewer than five words), the corpus comprises 5,038 reviews spanning 630 unique businesses across 23 business domains. This corpus is not used for training; it serves exclusively as the annotation target for the model-assisted pipeline described in Section 4.3.

Data collection and usage were conducted with explicit permission from the data owner, as noted in Section 7. The annotated corpus derived from these reviews is publicly released on Zenodo to facilitate further research on low-resource ABSA.[6] To support future domain-adaptive annotation work, we also release a domain-specific aspect subcategory taxonomy comprising 119 fine-grained sub-categories across 18 of the 23 business domains (e.g., *kredit*, *foiz_stavka*, *karta_xizmati* for banking; *shifokor_malakasi*, *diagnostika* for healthcare), available alongside the dataset on GitHub. This taxonomy is not used as the prediction label space in the current experiments; the fine-tuned models are evaluated using the original five UzABSA categories.

The domain distribution is highly skewed: *Restoran/Ovqatlanish* (restaurants and food service) is the largest domain with 1,626 reviews (32.3%), followed by *Bank/Moliya* (banking and finance) with 577 reviews (11.5%). The ten most frequent domains together account for approximately 89.9% of the corpus; Table 2 lists their counts and proportions.

The reviews are short, averaging 13.55 words and 96.26 characters. They are also complete reviews, and therefore roughly twice the length of the UzABSA training instances, which average 6.42 words. The training instances consist of annotation spans at the level of clauses or sentences, not of complete reviews. This means that the annotation model has to cope with the differences in length and in the level of detail at the inference stage. The corpus is predominantly Latin-script: 92.8% of reviews are in Uzbek Latin, 5.2% in Russian Cyrillic, and 2.0% in Uzbek Cyrillic. This is consistent with the process of script transition after the Soviet times and the presence of a Russian-speaking community among Uzbek internet users. No script normalisation or transliteration has been performed before annotation.

5 https://sharh.commeta.uz/en
6 https://doi.org/10.5281/zenodo.22146677

Table 2. Top-10 domain distribution in the multi-domain Uzbek review corpus (5,038 reviews total).

Rank	Domain	Reviews	%
1	Restoran/Ovqatlanish (Restaurant/Dining)	1,626	32.3
2	Bank/Moliya (Banking/Finance)	577	11.5
3	Boshqa (Other)	507	10.1
4	To'lov tizimlari (Payment systems)	328	6.5
5	Elektron tijorat (E-commerce)	296	5.9
6	Ta'lim (Education)	287	5.7
7	Tibbiyot/Sog'liqni saqlash (Healthcare/Medicine)	285	5.7
8	Telekommunikatsiya (Telecommunications)	234	4.6
9	Oziq-ovqat do'konlari (Grocery stores)	215	4.3
10	Gul/Sovg'a (Flowers/Gifts)	173	3.4
	Remaining 13 domains	510	10.1
	Total	**5,038**	**100.0**

3.2. Task Formulation

We cast ABSA as conditional text generation rather than separate sequence labeling/classification. Given an Uzbek review, the model produces a single structured JSON containing all aspect terms with their categories and sentiment polarities, avoiding multi-stage pipelines, reducing error propagation, and enabling end-to-end optimization.

Formally, let x denote the instruction-formatted input (system prompt and review text) and let $y = (y_1, \ldots, y_T)$ be the tokenized JSON annotation, where each aspect entry is a triple (t_j, c_j, p_j) of term, category, and polarity. Fine-tuning maximizes the conditional log-likelihood of the target sequence,

$$\mathcal{L}(\theta) = -\sum_{(x,y)\in\mathcal{D}} \sum_{t=1}^{T} \log p_\theta\big(y_t \,\big|\, y_{<t}, x\big), \tag{1}$$

where θ are the trainable adapter parameters (Section 4.1) and $\mathcal{D}$ is the instruction-formatted training set. At inference time the model generates $\hat{y}$ autoregressively with near-greedy decoding, and the aspect set $\{(\hat{t}_j, \hat{c}_j, \hat{p}_j)\}$ is recovered by JSON parsing.

We use a ChatML-style instruction template with system/user/assistant roles: the system sets Uzbek-ABSA expertise, the user supplies the review and extraction directives, and the assistant returns the JSON annotation. The full formatting is shown in Figure 1.

The JSON output format serves multiple methodological purposes: (a) it ensures structured, parseable responses that enable automatic metric computation without post-processing; (b) it naturally accommodates variable numbers of aspects per review; and (c) it facilitates batch annotation pipelines for large-scale corpus creation.

Two points about the target deserve emphasis, because they define what the reported metrics do and do not measure. First, UzABSA annotates aspect terms and aspect categories as *independent* layers: a category label is not attached to a particular term span. We preserve that independence rather than heuristically pairing the layers, so a target list contains term records (`term`, `polarity`) alongside category records (`category`, `polarity`), and the model learns to emit both. Consequently the "pair F1" reported throughout is an *aspect–polarity* pair, not a term–category–polarity triple, and category-layer accuracy is not scored here; constructing genuine triples would require a fresh annotation pass and is left for future work. Second, the four source polarity labels include *conflict*, which accounts for only 1.4% of term annotations (103 of 7,412). Our preprocessing retains only *positive*, *negative*, and *neutral*, so both training and evaluation—including the macro-averaged sentiment F1—operate over three classes.

```
<|im_start|>system
Siz o'zbek tilida matnlardan aspektlarni va ularning hissiyotlarini aniqlash bo'yicha mutaxassissiz.
Berilgan matndan barcha aspekt terminlarini, ularning kategoriyalarini va hissiyot polaritesini
(positive, negative, neutral, conflict) ajratib oling.
Javobni quyidagi Python dictionary formatida qaytaring:
{ "aspects": [ { "term": "...", "category": "...", "polarity": "..." } ] }
```

```
<|im_start|>user
Quyidagi o'zbek tilidagi matndan aspektlarni, kategoriyalarni va sentimentini aniqlang:
Matn: "Bu restoranning ovqatlari juda mazali, lekin narxlari qimmat."
```

```
<|im_start|>assistant
{"aspects": [
{"term": "ovqatlari", "polarity": "positive"},
{"term": "narxlari", "polarity": "negative"},
{"category": "ovqat", "polarity": "positive"},
{"category": "narx", "polarity": "negative"}
]}
```

Figure 1. ChatML instruction template for Uzbek ABSA. The target is a single JSON list carrying *two* annotation layers, exactly as they appear in the SemEval-style source. A *term-level* record names a phrase that literally occurs in the review text and gives its polarity (*ovqatlari* "food", *narxlari* "prices"). A *category-level* record instead names one of five fixed aspect categories and gives its polarity (*ovqat*, *narx*). The two layers are annotated independently in UzABSA: a category is not attached to any particular term span, so the layers are not paired into {term, category, polarity} triples. This study evaluates the term layer (Section 4.4).

This generative formulation lets the model identify aspect boundaries (the exact phrase in the text), assign it to a category, and classify its polarity in one pass, rather than as separate sequential steps. The instruction text is provided entirely in Uzbek to maintain language consistency and reduce code-switching effects that could degrade model performance on this low-resource language.

4. Methodology

Our design requires models with (i) openly available (open-weight) checkpoints, (ii) instruction-tuned variants for complex prompt following, (iii) ∼7–8B parameters for controlled comparison under a fixed compute budget, and (iv) 4-bit quantized availability compatible with QLoRA. Under these constraints, we select three complementary LLMs: Qwen 2.5-7B-Instruct [2], Llama 3.1-8B-Instruct [3], and DeepSeek-R1-Distill-Qwen-7B (a Qwen-7B backbone distilled from DeepSeek-R1 [4]).

Qwen 2.5-7B-Instruct [2] is a 7.6B decoder-only transformer with multilingual pre-training (reported >29 languages) and a GQA+SwiGLU+RoPE configuration (28 heads, 4 KV heads), which may benefit Uzbek. Llama 3.1-8B-Instruct [3] is an 8B GQA+SwiGLU+RoPE model (32 heads, 8 KV heads) with different depth/width (32 layers; 4,096 hidden vs. Qwen's 28 layers; 3,584), and its Turkic coverage is less explicitly documented. DeepSeek-R1-Distill-Qwen-7B matches Qwen architecturally but differs in weights due to reasoning-oriented distillation rather than standard instruction tuning [4]. This choice supports two informative contrasts under a single fine-tuning protocol: Qwen vs. Llama, and Qwen vs. its R1-distilled sibling, the latter holding the architecture fixed. We stress that neither contrast isolates a single causal factor. The checkpoints differ in several respects at once: tokenizer, pre-training corpus, instruction-tuning recipe, and, for Llama, parameter count. Any observed difference is therefore attributable to the checkpoints as wholes, not to architecture or distillation alone. All models are loaded from HuggingFace Hub in pre-quantized 4-bit NormalFloat via Unsloth,[7] avoiding quantization as a confound.

4.1. QLoRA Fine-Tuning

We adopt QLoRA [5] for parameter-efficient adaptation. LoRA [21] freezes the pre-trained weights and injects a trainable low-rank decomposition into each adapted linear layer. For a frozen projection $W_0 \in \mathbb{R}^{d\times k}$, the adapted forward pass becomes

$$h \;=\; W_0 x \;+\; \Delta W x \;=\; W_0 x \;+\; \frac{\alpha}{r}\, BAx, \qquad A \in \mathbb{R}^{r\times k},\; B \in \mathbb{R}^{d\times r}, \tag{2}$$

where $r \ll \min(d,k)$ is the adapter rank and α a scaling constant; only A and B are updated during training. QLoRA additionally stores W_0 in 4-bit NormalFloat (NF4) precision with double quantization, so the forward pass computes

$$h \;=\; \mathrm{dequant}\big(W_0^{\mathrm{NF4}}\big)\, x \;+\; \frac{\alpha}{r}\, BAx \tag{3}$$

in BFloat16, reducing the memory footprint of a 7B-parameter model to under 6 GB while leaving gradient flow through the full-precision adapters intact [5].

In our configuration, adapters (r=16, α=32, dropout 0.05, no bias) are applied to all attention and feed-forward linear projections. The resulting adapter size depends on the backbone's depth and width: 40.4M trainable parameters for the two Qwen-architecture models (28 layers, 3,584 hidden) and 41.9M for Llama 3.1-8B (32 layers, 4,096 hidden), corresponding to roughly 0.53% and 0.52% of the respective nominal parameter counts. Training is performed via the SFTTrainer with Unsloth's optimized fused kernels and gradient checkpointing to minimize memory footprint. All three models are trained under strictly identical hyperparameters: 8-bit AdamW optimizer, learning rate of 2×10^{-4} with a cosine schedule and 0.1 warmup ratio, 0.01 weight decay, and gradient clipping at 1.0. We train for 1,000 steps on the 5,480-example split using a maximum sequence length of 2,048 tokens and an effective batch size of 16 (per-device batch 4, 4 accumulation steps).

4.2. Encoder Baselines

To situate the generative formulation against classical ABSA architectures, we fine-tune two monolingual Uzbek encoders as token-classification baselines: BERTbek [39], a BERT-base model pre-trained on Uzbek news text, and TahrirchiBERT [40], a RoBERTa-base model pre-trained on approximately 5B tokens of Uzbek Latin-script text. Both have ∼110M parameters—roughly 1.5% of the size of the 7–8B LLMs.

Joint extraction and polarity classification is cast as BIO tagging with polarity-augmented labels (`O`, `B-`/`I-`{*positive, negative, neutral, conflict*}), so a single encoder predicts aspect spans and their sentiment simultaneously; decoded spans are converted to the same {term, polarity} structure used by the generative models, enabling identical scoring. Gold character spans are projected onto whitespace/punctuation word tokens; 98.2% of training and 98.5% of validation aspect terms could be localized this way (unlocalizable terms are dropped from encoder *training* only—evaluation for all systems uses the full reference set). Both encoders are trained on the identical 5,480-example split used for the LLMs (learning rate 3×10^{-5}, 4 epochs, batch size 16, maximum length 192, seed 42) and evaluated on the same 609-example set. Notably, the encoder baselines require no GPU-server infrastructure: each fine-tuning run completes in approximately 10 minutes on an Apple M2 laptop.

[7] https://github.com/unslothai/unsloth

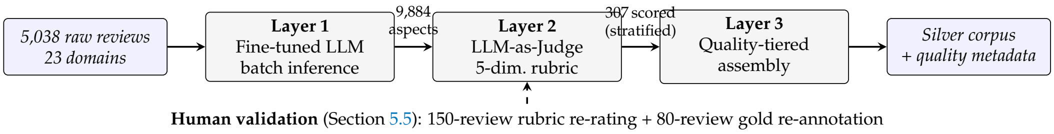

Figure 2. The three-layer model-assisted annotation pipeline. A locally fine-tuned LLM annotates the full multi-domain corpus; an external LLM judge scores a stratified sample on a five-dimension rubric; assembly partitions the corpus into quality tiers. A native-speaker validation study (dashed) calibrates the judge and contributes a reconciled double-annotated subset.

Algorithm 1 Three-layer model-assisted annotation with quality tiering

Require: reviews $\mathcal{R} = \{x_1, \ldots, x_N\}$ with domain labels d_i; fine-tuned annotator M; judge J; sample size m; thresholds $\tau_{\text{inc}} = 3.5$, $\tau_{\text{exc}} = 2.5$

Ensure: quality-tiered corpus $\mathcal{C}$

Layer 1 (annotation): for each $x_i \in \mathcal{R}$: $\hat{y}_i \leftarrow \text{parse}(M(x_i))$ ▷ greedy decoding; checkpoint every 100 reviews

Layer 2 (judging): draw stratified sample $\mathcal{S} \subset \mathcal{R}$, $|\mathcal{S}| = m$, covering every domain d

for $x_i \in \mathcal{S}$ **do**

$(s_i^{\text{comp}}, s_i^{\text{acc}}, s_i^{\text{sent}}, s_i^{\text{rel}}, s_i^{\text{ov}}) \leftarrow J(x_i, \hat{y}_i)$ ▷ 1–5 Likert per dimension

end for

Layer 3 (assembly): for each $x_i \in \mathcal{R}$:

$$\text{tier}(x_i) \leftarrow \begin{cases} \textit{Include} & x_i \in \mathcal{S} \wedge s_i^{\text{ov}} \geq \tau_{\text{inc}} \\ \textit{Flag} & x_i \in \mathcal{S} \wedge \tau_{\text{exc}} \leq s_i^{\text{ov}} < \tau_{\text{inc}} \\ \textit{Exclude} & x_i \in \mathcal{S} \wedge s_i^{\text{ov}} < \tau_{\text{exc}} \\ \textit{Unjudged} & x_i \notin \mathcal{S} \end{cases}$$

return $\mathcal{C} = \{(x_i, \hat{y}_i, d_i, \text{tier}(x_i), \text{scores if judged})\}$

4.3. Multi-Domain Annotation Pipeline

To build a silver-standard, multi-domain Uzbek ABSA dataset, we design a three-layer pipeline that combines local inference with external quality validation. This addresses a common low-resource constraint: for the 23 out-of-domain business categories, no gold annotations exist and large-scale Uzbek expert labeling is prohibitively costly. Figure 2 gives a schematic overview, and Algorithm 1 states the procedure precisely.

Layer 1: Model Inference. Qwen 2.5-7B serves as the annotation engine, selected from the three-model comparison (Section 5.1) for operational reliability: a 100% JSON parse rate, the highest ATE recall (maximising aspect coverage), and the fastest inference throughput.[8] We run greedy batch inference over all 5,038 reviews using the same instruction template and hyperparameters as evaluation, with checkpoint-based crash recovery (state saved every 100 reviews).

Layer 2: LLM-as-Judge Quality Scoring. Without references, we use GPT-4o-mini (via the OpenAI API) as an external judge. We draw a stratified sample of 307 reviews across all 23 domains to cover both high-frequency domains (e.g., restaurants, n=50) and low-frequency domains (e.g., insurance, n=3). For each sampled review, the judge assigns 1–5 Likert scores for five rubric dimensions: completeness (coverage of expressed opinions), accuracy (terms present in the text), sentiment correctness (polarity labels), relevance (category–domain fit), and overall quality. The judge relies only on Uzbek text comprehension, consistent with the reference-free LLM-as-Judge paradigm in Section 2. Judging used OpenAI's `gpt-4o-mini` (accessed via API on 25 February 2026) with temperature 0.1,

[8] The case study was executed with the Qwen fine-tune from an earlier revision of the preprocessing pipeline (before the split and prompt-serialization fixes of Section 5.3). Because annotation quality is assessed directly by the judge layer and the human validation study, the pipeline's conclusions are unaffected; see Section 6.

a 300-token response cap, and one call per review; all 307 responses parsed successfully, and the full rubric prompt is released with the code.

Layer 3: Quality-Tiered Assembly. We partition judged outputs into *Include* (overall ≥ 3.5), *Flag for review* ($2.5 \leq$ overall < 3.5), and *Exclude* (overall < 2.5). The assembly script joins predictions with quality metadata (per-dimension judge scores and business category) and exports: (i) full JSON with provenance metadata, (ii) a silver-standard subset comprising the judge-included reviews together with the non-sampled remainder, and (iii) JSONL for direct loading via HuggingFace `datasets`. Individual quality tiers exist only for the 307-review judged sample; every non-sampled review carries an explicit "unjudged" flag rather than a tier. Because the stratified sample deliberately over-represents low-frequency domains, its raw 63.5% inclusion rate is a sample-level figure; reweighting the per-domain inclusion rates by the corpus domain distribution yields an estimated corpus-wide inclusion rate of 67.3% (95% CI [62.1%, 72.6%]). The resulting corpus is released on Zenodo (DOI: 10.5281/zenodo.22146677).

4.4. Evaluation Framework

We evaluate model performance along four complementary dimensions, each targeting a distinct facet of the ABSA generation task. Throughout, let G_i and P_i denote the gold and predicted aspect-term sets for review i, over an evaluation set of n reviews.

Aspect Term Extraction (ATE). We report micro-averaged precision, recall, and F1 under two matching criteria. *Exact match* requires the predicted term to be identical to the gold term after case-folding and whitespace normalisation. *Partial match* relaxes this constraint: a prediction is counted as correct if the predicted string is a substring of the gold term or vice versa (i.e., a superstring relationship), accommodating morphological variation in Uzbek—for example, matching *ovqatlari* against *ovqat*. Formally, with matching predicate $\mu \in \{\mu_{\text{ex}}, \mu_{\text{par}}\}$,

$$\mu_{\text{ex}}(t,g) = \mathbb{1}\left[\text{norm}(t) = \text{norm}(g)\right], \tag{4}$$
$$\mu_{\text{par}}(t,g) = \mathbb{1}\left[\text{norm}(t) \sqsubseteq \text{norm}(g) \vee \text{norm}(g) \sqsubseteq \text{norm}(t)\right], \tag{5}$$

where $\sqsubseteq$ denotes the substring relation, the micro-averaged scores are

$$\mathrm{P}_\mu = \frac{\sum_{i=1}^{n} |\{\, t \in P_i : \exists\, g \in G_i,\ \mu(t,g)\,\}|}{\sum_{i=1}^{n} |P_i|}, \qquad \mathrm{R}_\mu = \frac{\sum_{i=1}^{n} |\{\, g \in G_i : \exists\, t \in P_i,\ \mu(t,g)\,\}|}{\sum_{i=1}^{n} |G_i|}, \tag{6}$$

and $\mathrm{F1}_\mu = 2\,\mathrm{P}_\mu \mathrm{R}_\mu / (\mathrm{P}_\mu + \mathrm{R}_\mu)$. Micro-averaging across all test examples avoids bias from examples with few or many aspect terms. Two implementation properties should be noted: terms are normalized (lower-cased, whitespace-trimmed) and compared as sets, so repeated identical terms within one review are collapsed; and the existential predicates above do not enforce a one-to-one prediction–gold alignment for partial matches, so an optimal bipartite matching over character offsets would be a stricter variant, which we leave for future work. Both properties apply identically to every system evaluated, so cross-system comparisons are unaffected.

End-to-End ABSA (E2E). The primary evaluation metric is the *aspect–polarity pair F1*, which requires both the aspect term and its sentiment polarity to be correct for a true positive: Equation (6) is applied to the tuple sets $\{(t,p) : t \in P_i\}$ and $\{(g,p^*) : g \in G_i\}$ with the joint predicate $\mu_{\text{ex}}(t,g) \wedge \mathbb{1}[p = p^*]$. This metric captures joint extraction and classification performance in a single score and is directly comparable across models.

Sentiment Classification. For aspect terms that are correctly matched (exact match), we additionally report sentiment accuracy and macro-averaged F1 over the K=3 polarity classes retained after preprocessing,

$$\text{Acc} = \frac{|\{\text{matched terms with } p = p^*\}|}{|\{\text{matched terms}\}|}, \qquad \text{macro-F1} = \frac{1}{K}\sum_{k=1}^{K} \text{F1}_k. \tag{7}$$

Because the polarity distribution is skewed (Section 3), macro-F1 weights the rarer *negative* and *neutral* classes equally with the dominant *positive* class (*conflict* is excluded in preprocessing, Section 3.2). This isolates the model's sentiment assignment capability from its term extraction performance.

Instruction Following. We measure the *JSON parse success rate*—the percentage of model outputs that yield a syntactically valid JSON object containing the expected `aspects` key. When a generation fails to parse, the evaluator falls back to a regular-expression salvage pass and scores whatever aspects it recovers, which is frequently none; failed parses are therefore counted as (usually empty) predictions rather than discarded. This is the conservative choice—it charges the model for its formatting failures as false negatives—but it means the parse rate and the ABSA metrics are not independent, and the encoders, which emit structured output by construction, are not exposed to this penalty at all.

Statistical Significance. Where two systems A and B are compared on the same evaluation set, we assess whether the observed metric difference $\hat{\delta} = m(A) - m(B)$ is distinguishable from zero using a paired bootstrap over reviews [44]: for $b = 1, \ldots, B$ resamples (B = 10,000), we draw n reviews with replacement, recompute the metric for both systems on the identical resample, and record δ_b. We report the 95% percentile confidence interval of $\{\delta_b\}$ and the two-sided p-value

$$p \;=\; 2\min\Big(\tfrac{1}{B}\textstyle\sum_b \mathbb{1}[\delta_b \le 0],\; \tfrac{1}{B}\sum_b \mathbb{1}[\delta_b \ge 0]\Big). \tag{8}$$

Pairing by review preserves the per-example correlation between systems and therefore tests the difference rather than the two absolute scores. All p-values are reported unadjusted. As a guard against multiple-comparison effects, we also apply a Holm–Bonferroni correction over the family of 28 post-hoc system-comparison tests (seven system pairs $\times$ four metrics): only the Llama–DeepSeek pair-F1 difference remains significant after adjustment (raw p=0.0016; corrected p=0.045), together with the three fine-tuned-versus-zero-shot contrasts (all raw p<0.001), which survive any standard correction. Individual sub-0.05 results elsewhere in Section 5 should therefore be read as exploratory evidence rather than confirmed differences.

All inference is performed with greedy decoding (`do_sample=False`, `max_new_tokens=512`), which is deterministic and reproducible. (The code additionally sets `temperature=0.1`, but this has no effect when sampling is disabled.) For the multi-domain annotation quality assessment, we supplement automatic metrics with the five-dimension LLM-as-Judge rubric described in Section 4.3.

5. Results

All experiments run on NVIDIA RTX A6000 GPUs (48 GB VRAM) using Python 3.13 with Unsloth, TRL, PEFT, and bitsandbytes; each run occupies a single GPU. The UzABSA dataset is split 90/10 into training (5,480) and evaluation (609) sets with a fixed seed of 42. The split is *grouped by review text*: all annotation records sharing a normalised review string are assigned to the same side, so no review can appear on both sides of the partition. This matters because the corpus annotates aspect terms and aspect categories as separate records (Section 3.2) and because short formulaic reviews recur verbatim, so a naive record-level

Table 3. Training parameters were examined on a single RTX A6000. All experiments were conducted with the same set of QLoRA hyperparameters; the size of adapters depended on the structure of the model—the Qwen-based models used 40.4 million trainable parameters while Llama 3.1-8B used 41.9 million. Wall clock time is provided for the only experiment that was carried out continuously from start to finish. The other two experiments were restored using the optimizer checkpoint after a power failure; therefore, their wall clock time and throughput are not efficient metrics(*).

Metric	Qwen 2.5-7B	Llama 3.1-8B	DeepSeek-R1-7B
Compute time (min)	—*	—*	84
GPU mem. allocated (GB)	5.4	5.6	5.5
Total FLOPs	2.19×10^{17}	2.49×10^{17}	2.37×10^{17}
Best eval loss (step)	0.1726 (700)	**0.1476** (1000)	0.1744 (1000)

Table 4. ABSA evaluation on the 609-example deduplicated evaluation set. Best scores per metric are **bold**. LLMs use greedy decoding; encoder baselines (BERTbek, TahrirchiBERT) use BIO-polarity token classification (Section 4.2) and produce structured output by construction, so the JSON parse rate does not apply. Per-system percentile-bootstrap 95% confidence intervals for every score in this table are reported in Table A2 (Section B); their half-widths range from ±0.023 to ±0.037, so most inter-system gaps are not statistically significant (see text and Section 5.2).

		Fine-tuned LLMs (QLoRA, 7–8B)			Encoders (∼110M)	
Task	**Metric**	**Qwen 2.5**	**Llama 3.1**	**DeepSeek-R1**	**BERTbek**	**Tahrirchi**
ATE (exact)	Precision	0.7141	0.7185	0.7143	0.6802	**0.7287**
	Recall	0.7015	0.6893	0.6445	**0.7042**	0.6744
	F1	**0.7077**	0.7036	0.6776	0.6920	0.7005
ATE (partial)	Precision	0.8094	0.8175	**0.8301**	0.7864	0.8182
	Recall	0.7951	0.7843	0.7490	**0.8141**	0.7571
	F1	**0.8022**	0.8006	0.7874	0.8000	0.7865
E2E Pair	Precision	0.6506	**0.6648**	0.6361	0.5976	0.6471
	Recall	**0.6391**	0.6377	0.5739	0.6187	0.5997
	F1	0.6448	**0.6510**	0.6034	0.6080	0.6225
Sentiment	Accuracy	0.9110	**0.9252**	0.8905	0.8786	0.8893
	Macro-F1	0.8622	**0.8867**	0.8314	0.8126	0.8140
JSON parse	Success rate	**100.0%**	97.4%	96.9%	—	—

random split leaks reviews across the partition; Section 5.3 quantifies what that leakage does to the reported numbers. Each of the three models is trained for exactly 1,000 gradient steps under the QLoRA configuration described in Section 4.1; the *only* variable across runs is the base model. We use the term evaluation set rather than test set because the same partition is monitored during training; no hyperparameter search was performed on it after observing downstream ABSA metrics. The fine-tuned encoder baselines (Section 4.2) and the unadapted zero-shot baselines (Section 5.2) are evaluated on this identical partition, so all systems in Table 4 are directly comparable. Table 3 summarises the resulting training efficiency.

5.1. Three-Model ABSA Comparison

Table 4 presents the full evaluation on the 609-example evaluation set, covering the three QLoRA-fine-tuned LLMs and the two fine-tuned Uzbek encoder baselines (Section 4.2). Four complementary findings emerge.

Qwen leads extraction and is the most reliable formatter. The Qwen-2.5-7B model obtains the highest exact ATE F1 (0.7077) and partial ATE F1 (0.8022) scores, as well as a JSON parsing rate of 100%, which is a key strength for real-world annotation tasks. The replicate of Qwen-2.5-7B, with the only difference being the seed used, obtains a score of 0.7101/0.6554 (exact-ATE/pair F1) against 0.7077/0.6448 for the original model. Thus, differences of about ±0.01 at the seed level should be

Llama leads on sentiment and end-to-end performance—but no significant difference from Qwen is detected. Llama 3.1-8B obtains the highest E2E pair F1 (0.6510), sentiment accuracy (0.9252), and sentiment macro-F1 (0.8867) of any system, with precision-oriented extraction compensating for slightly lower recall. However, a paired bootstrap over reviews (B=10,000; Equation (8)) finds no significant Qwen–Llama difference on any headline metric (exact-ATE $\Delta=+0.004$, p=0.79; pair F1 $\Delta=-0.006$, p=0.68; sentiment accuracy $\Delta=-0.014$, p=0.27): the observed gaps are within seed-level and sampling variation—though absence of significance is not evidence of equivalence—and the choice between the two models can reasonably rest on secondary criteria such as parse reliability, licence, and tokenizer efficiency.

DeepSeek trails despite reasoning distillation. DeepSeek-R1-Distill-Qwen-7B scores lowest among the LLMs on every F1 metric (exact ATE 0.6776; pair F1 0.6034, below both Qwen, p=0.009, and Llama, p=0.002, under the paired bootstrap; the gap to Llama is the only pairwise difference that survives the Holm correction of Section 4.4). Its failure mode is recall: it finds the fewest gold aspects (0.6445 exact-ATE recall) while its precision matches the other two, and it under-predicts (665 predictions vs. 737 gold terms; ratio 0.90, versus 0.98 for Qwen and 0.96 for Llama). The R1 reasoning distillation, designed for chain-of-thought tasks, confers no advantage on direct structured extraction here—a pattern the zero-shot controls in Section 5.2 make considerably more vivid.

Encoders are competitive at extraction; LLMs win on sentiment. The fine-tuned Uzbek encoders remain strong extraction baselines at $\sim$1.5% of the LLMs' parameters: TahrirchiBERT's exact-ATE F1 (0.7005) sits within 0.007 of Qwen's, and the paired bootstrap finds no significant extraction difference between the best LLM and either encoder (Qwen vs. TahrirchiBERT exact-ATE p=0.61; vs. BERTbek p=0.29), nor between the two encoders on any metric (exact-ATE p=0.46, partial-ATE p=0.13, pair F1 p=0.25, sentiment p=0.39). The joint-prediction advantage, by contrast, is real but uneven: Qwen's pair-F1 and sentiment margins over BERTbek are significant (p=0.020 and p=0.021), though not over TahrirchiBERT (p=0.13 and p=0.14), while Llama's sentiment-accuracy margin holds even against the stronger encoder ($\Delta=+0.036$ over TahrirchiBERT, p=0.014). These sub-0.05 joint-prediction results are unadjusted and do not survive the Holm correction of Section 4.4, so we read them as consistent exploratory evidence of an LLM advantage rather than individually confirmed differences. The LLM advantage concentrates elsewhere: sentiment macro-F1 (Llama 0.8867, Qwen 0.8622 vs. $\approx$0.81 for both encoders) and end-to-end pair F1 (0.6510/0.6448 vs. 0.6080/0.6225). For pure span extraction in Uzbek, a laptop-trainable 110M encoder therefore remains a highly competitive choice, whereas the generative LLMs deliver their value in joint aspect–sentiment prediction, categorical labeling within the same output structure, and the instruction-following flexibility that the annotation pipeline in Section 4.3 depends on.

Training Loss vs. Task Performance. Consistent with our second research question, ranking the models by validation loss, as visualized in Fig. 3, diverges from ranking them by downstream task metrics. Llama achieves the lowest evaluation loss (0.1476) and the highest E2E pair F1, yet Qwen surpasses it on exact ATE F1 despite a 17% higher evaluation loss (0.1726). More strikingly, Qwen and DeepSeek finish with nearly indistinguishable evaluation losses (0.1726 vs. 0.1744, a 1% gap) while differing by 3 points of exact-ATE F1 and 4 points of pair F1. Cross-entropy loss captures token-level prediction quality but is blind to structured output correctness—aspect boundary precision, JSON formatting, and

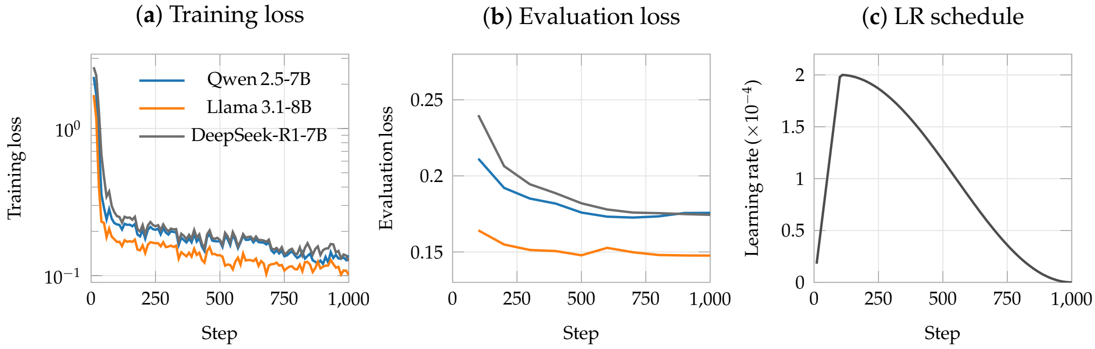

Figure 3. Training dynamics of the three QLoRA fine-tuning runs over 1,000 gradient steps on the deduplicated split, plotted from the released per-step training logs. (**a**) Training loss (log scale, logged every 10 steps). (**b**) Evaluation loss on the 609-example set (every 100 steps): Llama 3.1-8B reaches the lowest loss (0.148), while Qwen 2.5-7B (0.173, minimum at step 700) and DeepSeek-R1-7B (0.174) are nearly indistinguishable—despite clearly different downstream ABSA scores (Table 4). (**c**) The cosine learning-rate schedule with 10% warmup shared by all runs.

Table 5. Zero-shot (ZS) versus QLoRA fine-tuned (FT) performance, same split, prompt, and decoding. Ratio is total predicted aspect terms over gold terms (737); values above 1 indicate over-prediction. Fine-tuning at least doubles exact-ATE F1 and triples pair F1 for every architecture; all FT–ZS differences are significant at $p < 0.001$ (paired bootstrap), surviving the Holm correction of Section 4.4.

Model		**ATE F1 (ex.)**	**ATE F1 (par.)**	**Pair F1**	**Sent. macro-F1**	**JSON parse**	**Ratio**
Qwen 2.5-7B	ZS	0.3222	0.5593	0.2100	0.4719	99.8%	1.80
	FT	**0.7077**	**0.8022**	0.6448	0.8622	**100.0%**	0.98
Llama 3.1-8B	ZS	0.2843	0.4924	0.1999	0.6280	95.4%	1.67
	FT	0.7036	0.8006	**0.6510**	**0.8867**	97.4%	0.96
DeepSeek-R1-7B	ZS	0.2169	0.3182	0.1540	0.5683	46.3%	1.10
	FT	0.6776	0.7874	0.6034	0.8314	96.9%	0.90

polarity alignment. Model selection for structured extraction tasks should therefore not rely on validation loss alone.

5.2. What Does Fine-Tuning Buy? Zero-Shot Controls

To separate what QLoRA contributes from what the instruction-tuned base models already know, we evaluate each *unadapted* base model on the identical 609-example split with the identical Uzbek prompt and decoding settings. Table 5 reports the comparison.

Three observations follow. First, the fine-tuning effect is large and uniform: exact-ATE F1 rises by 0.38–0.46 and pair F1 by 0.43–0.45 absolute for every architecture, so the headline numbers in Table 4 are overwhelmingly a product of adaptation rather than of latent Uzbek ABSA competence in the base models. Second, fine-tuning buys *task* competence, not *format* compliance: Qwen already emits parseable JSON for 99.8% of zero-shot outputs, but attaches it to nearly twice as many aspect terms as the gold standard (ratio 1.80) with poorly calibrated polarities (sentiment macro-F1 0.4719). Fine-tuning's clearest behavioural signature is this calibration of *how much* to extract—the prediction ratio moves from 1.10–1.80 to 0.90–0.98. Third, DeepSeek is the exception that proves the format rule: zero-shot, its R1-style reasoning traces break JSON parsing on more than half the examples (46.3%), and fine-tuning is what suppresses the reasoning preamble (96.9%)—yet even with formatting repaired, it remains the weakest of the three, indicating that reasoning distillation neither helps unadapted structured extraction nor survives as an advantage after adaptation.

Table 6. Effect of replacing the naive record-level random split with review-grouped, deduplication-aware splitting (identical sizes: 5,480/609). Encoders isolate the split effect; the LLM deltas additionally include a prompt-serialization fix in the same pipeline revision. Every system *improves* on the leak-free split.

System	ATE exact F1			Pair F1		
	naive	grouped	Δ	naive	grouped	Δ
Qwen 2.5-7B	0.6603	0.7077	+0.047	0.5795	0.6448	+0.065
Llama 3.1-8B	0.6549	0.7036	+0.049	0.5805	0.6510	+0.071
DeepSeek-R1-7B	0.6034	0.6776	+0.074	0.5018	0.6034	+0.102
BERTbek	0.6694	0.6920	+0.023	0.5615	0.6080	+0.047
TahrirchiBERT	0.6528	0.7005	+0.048	0.5505	0.6225	+0.072

Table 7. LLM-as-Judge aggregate quality scores (307-review stratified sample, 1–5 scale).

Dimension	Mean	Interpretation
Accuracy	4.47	Extracted terms are genuinely present
Sentiment	4.21	Polarity labels largely correct
Relevance	4.06	Categories contextually appropriate
Overall	3.75	Majority suitable for inclusion
Completeness	3.32	Main opinions captured; some missed

5.3. Effect of Deduplication-Aware Splitting

Short formulaic reviews (*mazali* "tasty", *ajoyib* "excellent") recur verbatim in the UzABSA corpus, and the corpus stores aspect terms and aspect categories as separate annotation records (Section 3.2). A naive random split over records therefore leaks: in an earlier revision of this study's pipeline, 61 of 609 evaluation items (10.0%) had an exact-text twin in training. Table 6 quantifies what review-grouped deduplication changes, using the encoder baselines—whose training pipeline is otherwise untouched by the fix—to isolate the split effect.

The direction of the change is instructive: contrary to the usual assumption that train–test leakage inflates scores, every system scores *higher* once leakage is removed. The mechanism lies in the record structure. Under the naive split, the term-layer and category-layer records of the same review could land on opposite sides, so a model repeatedly saw an eval-identical text in training paired with a *partial* target, and learned to reproduce exactly that partial answer at evaluation time. Deduplication also changes inferential conclusions, not just absolute scores: on the contaminated split, the two encoders' partial-ATE difference tested as significant (p=0.023), a finding that does not survive on the clean split (p=0.13). Leakage can thus manufacture spurious model rankings even while depressing every absolute number—a caution we believe generalizes to other low-resource benchmarks built from short user-generated text, where verbatim duplication is endemic.

5.4. Multi-Domain Quality Assessment

To evaluate cross-domain generalisation, we apply the annotation pipeline (Section 4.3) using the Qwen model—selected for its 100% parse rate and best extraction F1 among the LLMs—and score a stratified sample of 307 reviews via GPT-4o-mini as LLM-as-Judge. Table 7 reports the aggregate quality scores across five dimensions.

Quality-tier assignment using the overall score yields 63.5% *Include* ($\geq$3.5; n=195), 28.0% *Flag* (2.5–3.49; n=86), and 8.5% *Exclude* ($<$2.5; n=26). These proportions describe the stratified sample, which intentionally over-samples low-frequency domains; reweighted to the corpus domain distribution, the estimated corpus-wide inclusion rate is 67.3% (95% CI [62.1%, 72.6%]). The high accuracy score (4.47) confirms that the model rarely hallucinates

Table 8. Per-domain LLM-as-Judge quality scores (overall, 1–5 scale). Domains grouped by proximity to the training distribution.

Group	Domain	N	Comp.	Acc.	Sent.	Rel.	Overall
In-domain	Restoran	50	3.68	4.76	4.56	4.66	4.24
Near	Sayohat/Turizm	5	4.00	4.80	4.60	4.40	4.40
	Oziq-ovqat	15	3.47	4.33	4.47	4.13	3.80
Out-of-domain	Bank/Moliya	30	3.40	4.47	4.17	3.90	3.70
	Telekom.	25	3.04	4.36	3.64	3.64	3.32
	Ta'lim	20	2.90	4.20	3.80	3.45	3.25
Distant	Sug'urta	3	2.33	3.67	2.67	3.33	2.67
	Investitsiya	3	2.33	4.00	3.67	3.00	2.67

aspects; the lower completeness score (3.32) reflects a tendency to miss subsidiary or domain-specific opinions.

Domain-Level Quality Gradient. Table 8 presents per-domain overall scores for representative domains spanning the quality spectrum; complete scores for all 23 domains are provided in Table A1 (Section A). A clear gradient emerges from in-domain to out-of-domain categories: the model trained exclusively on restaurant reviews achieves its highest quality on *Restoran* (4.24) and near-domain categories sharing overlapping aspect vocabularies (e.g., *Sayohat/Turizm*, 4.40), while performance degrades on domains with specialised terminology such as *Telekommunikatsiya* (3.32) and *Ta'lim* (3.25). The most distant domains—*Sug'urta* (insurance) and *Investitsiya* (investment)—drop to 2.67, driven by completeness failures rather than accuracy: the accuracy sub-score remains $\geq$3.67 across all 23 domains. Overall, 15 of 23 domains (65%) meet the inclusion threshold ($\geq$3.5).

Figure 4 visualises the full 23-domain quality spectrum against the inclusion threshold, making the in-domain-to-distant gradient and the position of each domain relative to the threshold directly legible.

Annotation Pipeline Output. Batch annotation of the full 5,038-review corpus produces 9,884 aspects (average 1.96 per review) with a 100% JSON parse rate across all 23 domains. The polarity distribution is 75.1% positive, 16.5% negative, and 8.4% neutral—reflecting a natural skew in user-generated reviews. The aspect category distribution reveals a notable pattern: *boshqalar* (miscellaneous) dominates at 45.5%, followed by *ovqat* (17.9%), *xizmat* (15.9%), *muhit* (10.7%), and *narx* (9.0%).

5.5. Human Validation of Annotation Quality

Because both the silver annotations and their quality scores originate from models, we validate a subsample against native-speaker judgement. Two native Uzbek speakers—the second and third authors—annotated a stratified subsample of 150 reviews drawn from the 307 judged reviews (all 23 domains represented, deliberately oversampling distant and low-scoring domains), performing two tasks: (i) rating the model's predicted annotations on the same five-dimension 1–5 rubric used by the LLM judge, and (ii) re-annotating a subset of 80 reviews from scratch to obtain human gold references.

Inter-annotator agreement. Rubric ratings were produced after a joint calibration pass on the 1–5 scale. Table 9 reports agreement between the two raters on all 150 reviews, measured with quadratic-weighted Cohen's κ (which penalises large rating gaps more than adjacent ones) and Krippendorff's α on the interval scale. Agreement is high and consistent across dimensions ($\kappa = 0.911$–0.970, $\alpha = 0.911$–0.970), highest on sentiment correctness ($\kappa = 0.970$) and lowest on relevance ($\kappa = 0.911$); Spearman correlations between raters fall in the same band ($\rho = 0.920$–0.941). The two raters therefore apply the rubric consistently

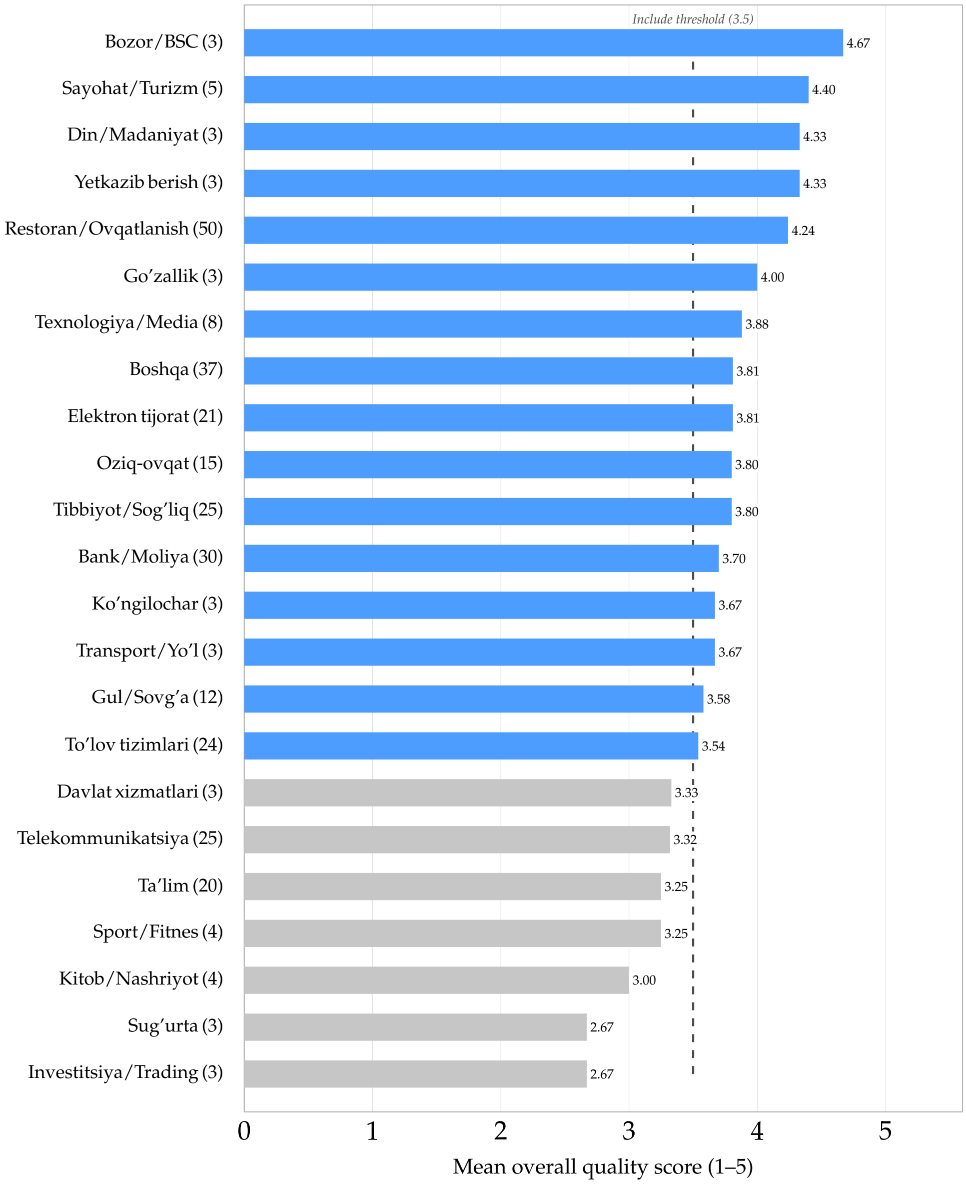

Figure 4. Mean LLM-as-Judge overall quality score for all 23 business domains (review counts in parentheses), sorted by score. Blue bars meet the inclusion threshold of 3.5 (dashed line); gray bars fall below it. The gradient runs from in-domain (*Restoran/Ovqatlanish*) and vocabulary-adjacent domains at the top to specialised financial domains (*Sug'urta*, *Investitsiya*) at the bottom.

enough for their mean rating to serve as the reference against which the LLM judge is calibrated below.

Gold annotation followed a two-stage protocol. In the first, fully independent pass, the two annotators' span-level agreement was low (exact-match F1 0.277, partial 0.415)—not because they disagreed about opinions, but because their segmentation and normalization conventions diverged (lemmatized vs. surface forms; naming the referent rather than the mention); polarity agreement on matched terms was perfect. In the second stage the annotators reconciled conventions and produced a consolidated reference with high residual span agreement (exact F1 0.927, polarity 100%), released as the reconciled double-annotated subset (`gold80`) alongside the silver corpus; because the reconciliation involved the same two annotators, we refrain from calling it fully adjudicated gold until an independent third annotator has resolved the residual disagreements. This two-stage outcome is itself

Table 9. Inter-annotator agreement between the two native-speaker raters on the 150-review rubric sample, per dimension. κ: quadratic-weighted Cohen's kappa; α: Krippendorff's alpha (interval); ρ: Spearman rank correlation.

Dimension	**Weighted κ**	**Krippendorff's α**	**Spearman ρ**
Sentiment	0.970	0.970	0.941
Accuracy	0.936	0.936	0.920
Completeness	0.930	0.930	0.937
Overall	0.923	0.922	0.923
Relevance	0.911	0.911	0.925

Table 10. Calibration of the GPT-4o-mini judge against the mean rating of two native-speaker annotators on the same 150 reviews (Spearman ρ, all $p < 10^{-6}$; MAE on the 1–5 scale).

Dimension	ρ	**MAE**	**Human mean**	**Judge mean**
Sentiment	0.806	0.34	4.25	4.24
Relevance	0.667	0.50	3.85	3.97
Overall	0.657	0.55	3.84	3.75
Accuracy	0.433	0.54	4.19	4.44
Completeness	0.462	0.94	4.04	3.30

informative: even for native speakers, *what counts as the aspect span* is the dominant source of ABSA annotation disagreement in Uzbek, while polarity is essentially unambiguous.

Judge calibration. Table 10 compares GPT-4o-mini's scores with the mean human rating on the same 150 reviews. Rank correlations are moderate to strong on the dimensions that drive dataset assembly—sentiment ($\rho = 0.81$), relevance ($\rho = 0.67$), and the tier-defining overall score ($\rho = 0.66$; all $p < 10^{-6}$)—and the judge's mean overall score (3.75) closely matches the human mean (3.84). Correlations are weaker for completeness and accuracy ($\rho \approx 0.43$–0.46), where the judge is systematically harsher on completeness (mean 3.30 vs. 4.04). We conclude that the LLM judge is a usable, though imperfect, proxy for native-speaker quality judgement, and that its quality tiers rank annotations in an order humans broadly agree with.

Model vs. human gold. The model gets F1 for exact match on the ATE task ranging between 0.209 and 0.224 and F1 for partial match from 0.425 to 0.433 (by annotator) when compared with the consolidated gold references, having perfect polarity agreement on exactly matching terms. There are two reference points that are useful to understand these results. First, the F1 partial match for the model (approximately 0.43) is consistent with the level reached by humans at level two before convention reconciliation (0.415), being substantially lower than the ceiling achieved via convention reconciliation (0.93). This difference is mostly due to the span segmentation problem, not to opinion detection or polarity. Secondly, when considering proximity domain groups, human measurement quality follows the gradient provided by the judge, dropping from 0.244 exact pair F1 on the general out-of-domain reviews to 0.170 for the farthest domains.

6. Discussion

The Completeness Bottleneck. Completeness (mean 3.32/5) is the worst quality dimension: the model successfully captures characteristics that are present in the text but sometimes misses domain-specific or subtle viewpoints, especially outside the restaurant domain. This is mainly due to the poor coverage of the training data, only five aspect categories, and the model has to map the unseen aspects to the generic *boshqalar* (miscellaneous) label, which is predominant with 45.5%. To solve this bottleneck, we would have to undertake domain-adaptive fine-tuning or expand the aspect taxonomy to better capture out-of-domain phenomena.

Checkpoint Effects. Distillation of R1 reasoning in the current context has not brought any benefit to structured ABSA. Even though Qwen-7B architecture is used as the basis of distilled checkpoint, the latter turned out to be inferior to both Qwen and Llama. The zero-shot controls can highlight parts of the mechanism: the checkpoints of the unadapted R1 model interfere with JSON parsing in more than half of the test cases (see Table Table 5). This implies that output tendencies related to reasoning do not work well within strict formats. Nevertheless, even after adapting the model to prevent preamble generation, there is no any effect observed. Llama has demonstrated the best results in terms of loss and sentiment accuracy but, on the other hand, Qwen is better at JSON formatting (100% parse rate and highest ATE recall).

Is a 7B LLM Worth It? In terms of the empirical results obtained for the encoder baselines, it becomes much clearer. There is no statistical significance difference compared to 7-8B large language models (LLMs) in aspect-term extraction, in case of 110M monolingual encoder trained in ten minutes on a laptop. This fact emphasizes that, for span identification in morphologically complex languages, targeted monolingual pre-training (like BERTbek, TahrirchiBERT) is still quite effective. The LLMs pay off the cost incurred in those cases when generation is truly joint and generative: there is around 5-7 F1 points improvement compared to encoders, the unified term- and category-level output during one generation process, and following natural language instructions for unseen domains – that corresponds to the goals of the multi-domain annotation pipeline development. Thus, in terms of Uzbek ABSA development teams with resource limitations, it might be recommended to use encoders for extraction tasks and leave the LLMs for those situations when generation is needed at the same time.

Loss as a Model Selection Proxy. Validation loss ranking diverges from ABSA performance: Llama's lowest loss does not translate to the best extraction, while Qwen outperforms on ATE F1 despite higher loss. Cross-entropy loss captures token-level likelihood but not structured-output quality such as JSON validity or aspect-boundary precision. This divergence is expected for structured-generation tasks, yet it is easy to overlook in practice; model selection for ABSA is therefore better guided by task-specific metrics than by validation loss alone.

Limitations. Several evaluation-design caveats remain. First, the 609-example split functions as a development set: evaluation loss on it is monitored during training (every 100 steps), and the same partition is then reused for the headline numbers, so the reported scores are optimistic relative to a genuinely untouched test set; a three-way train/development/test partition is the main methodological upgrade we would make with a larger corpus. Second, although the split is deduplicated by review text (Section 5.3) and now controlled against zero-shot baselines (Section 5.2), robustness to random seeds is only partially characterized: a single replicate (for Qwen) bounds seed variation at roughly ± 0.01 F1, but multi-seed replication of every system—which Table 6 suggests matters at exactly the scale of the observed inter-model gaps—remains undone. Third, few-shot prompting sits unexplored between the zero-shot controls and full fine-tuning, so we cannot say how much of the QLoRA gain cheaper in-context adaptation could recover. Finally, the multi-domain case study (Section 5.4) was produced with the annotation engine fine-tuned under the earlier revision of the preprocessing pipeline; since its output quality is assessed directly—by the LLM judge and the native-speaker validation study—the case-study conclusions are unaffected, but the released silver annotations do not benefit from the improvements in Table 6, and regenerating them with the corrected model is planned for the next dataset version. Tokenizer efficiency and larger-scale, convention-reconciled human verification remain for future work.

7. Conclusions

This paper presented is the first systematic study of parameter-efficient fine-tuning of open-weight LLMs for Uzbek ABSA. The findings show that QLoRA successfully performs joint extraction and classification in aspect-sentiment pairs through structured JSON generation, obtaining at least twice better extraction F1 and three times better end-to-end pair F1 against non-adapted zero-shot baselines. Moreover, fine-tuned LLMs achieve comparable performance to the fine-tuned Uzbek BERT encoders.

It should be noted that the architecture comparison has shown that Qwen 2.5-7B reliably performs aspect extraction and follows the instructions, while Llama 3.1-8B achieves the best precision of end-to-end sentiment matching, both models producing similar end-to-end F1. Also, validation loss appears to be an unreliable metric for the performance of downstream task in structured extraction.

In addition, the study provides evidence that a model trained only on restaurant reviews can produce useful annotations for further domains even though the quality of the annotations is getting noticeably worse with the growing distance between domains. The suggested three-stage pipeline, supplemented with a human-validated study of its outputs, produces a quality-scored multi-domain silver-standard dataset including a gold-standard subset, which represents a way of scalable dataset construction in low-resource settings. Future research includes developing a wider aspect taxonomy, domain-adaptive fine-tuning to improve the extraction completeness on different business domains, and using morphosyntactic structure provided by Uzbek UD treebanks and parsers[31–33] during extraction because span segmentation—not opinion detection—was the main point of disagreement between models and human annotators(Section 5.5).

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Data Availability Statement: The primary Uzbek aspect-based sentiment analysis dataset used for model fine-tuning is publicly available on Hugging Face at https://huggingface.co/datasets/Sanatbek/aspect-based-sentiment-analysis-uzbek. The multi-domain Uzbek business review corpus generated in this study, including the 80-review reconciled double-annotated subset (Section 5.5), is publicly available on Zenodo at https://zenodo.org/records/22146677 with DOI: 10.5281/zenodo.22146677. The released records contain the review text, business category and name, the star rating, and the annotation and quality metadata; contributor user names, review URLs, and posting timestamps present in the source platform are deliberately excluded. Code for the encoder baselines, the human-validation analysis, and the significance tests is available in the project repository.

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Abbreviations

The following abbreviations are used in this manuscript:

ABSA	Aspect-Based Sentiment Analysis
ACD	Aspect Category Detection
ASC	Aspect Sentiment Classification
ATE	Aspect Term Extraction
QLoRA	Quantized Low-Rank Adaptation

Appendix A. Full Per-Domain LLM-as-Judge Quality Scores

Table A1 reports LLM-as-Judge quality scores for all 23 business domains in the stratified evaluation sample (307 reviews total). The main text (Table 8) presents a selected subset grouped by training-distribution proximity; this table provides the complete picture, sorted by overall quality score in descending order. Scores are on a 1–5 Likert scale across five dimensions: Completeness, Accuracy (hallucination control), Sentiment Correctness, Relevance (category–domain fit), and Overall. Domains with $N \leq 5$ should be interpreted with caution due to small sample sizes. The "Include" threshold (Overall ≥ 3.5) is shown as a dashed reference.

Table A1. LLM-as-Judge (GPT-4o-mini) quality scores for all 23 business domains, sorted by Overall score (descending). Stratified evaluation sample: 307 reviews. Scale: 1–5 (higher is better). * Small sample ($N \leq 5$): interpret with caution.

Domain (Uzbek / English)	N	Comp.	Acc.	Sent.	Rel.	Overall
Bozor/BSC (Market/Shopping centre)*	3	3.67	5.00	5.00	5.00	4.67
Sayohat/Turizm (Tourism/Travel)*	5	4.00	4.80	4.60	4.40	4.40
Din/Madaniyat (Religion/Culture)*	3	4.33	5.00	4.00	5.00	4.33
Yetkazib berish (Delivery)*	3	3.67	5.00	4.67	4.33	4.33
Restoran/Ovqatlanish (Restaurant/Dining)	50	3.68	4.76	4.56	4.66	4.24
Go'zallik (Beauty services)*	3	3.33	4.67	5.00	4.00	4.00
Texnologiya/Media (Technology/Media)	8	2.88	4.50	5.00	3.88	3.88
Boshqa (Other/Miscellaneous)	37	3.43	4.54	3.89	4.38	3.81
Elektron tijorat (E-commerce)	21	3.43	4.48	4.10	4.14	3.81
Oziq-ovqat do'konlari (Grocery stores)	15	3.47	4.33	4.47	4.13	3.80
Tibbiyot/Sog'liqni saqlash (Healthcare)	25	3.40	4.60	4.20	4.36	3.80
Bank/Moliya (Banking/Finance)	30	3.40	4.47	4.17	3.90	3.70
Ko'ngilochar (Entertainment)*	3	3.33	4.67	4.33	4.00	3.67
Transport/Yo'l (Transport/Roads)*	3	3.33	4.00	4.00	4.00	3.67
Gul/Sovg'a (Flowers/Gifts)	12	3.08	4.08	4.83	3.67	3.58
To'lov tizimlari (Payment systems)	24	3.00	4.29	4.38	3.54	3.54
Below inclusion threshold (Overall < 3.5)						
Davlat xizmatlari (Government services)*	3	3.00	4.00	4.33	3.67	3.33
Telekommunikatsiya (Telecommunications)	25	3.04	4.36	3.64	3.64	3.32
Ta'lim (Education)	20	2.90	4.20	3.80	3.45	3.25
Sport/Fitnes (Gyms/Fitness)*	4	2.75	4.25	4.25	3.00	3.25
Kitob/Nashriyot (Books/Publishing)*	4	3.00	4.25	4.00	3.00	3.00
Sug'urta (Insurance)*	3	2.33	3.67	2.67	3.33	2.67
Investitsiya/Trading (Investment)*	3	2.33	4.00	3.67	3.00	2.67

Appendix B. Bootstrap Confidence Intervals for All Systems

Table A2 reports exact percentile-bootstrap 95% confidence intervals (B=10,000 resamples over the 609 evaluation reviews) for every score in Table 4. Half-widths range from ±0.023 (Llama sentiment accuracy) to ±0.037 (DeepSeek pair F1), and the intervals of the top-ranked systems overlap substantially on all four metrics—the reason the paired-bootstrap comparisons in Section 5.1 detect so few significant differences. Note that these are per-system intervals on absolute scores; because the systems are evaluated on identical reviews, the *paired* tests in Section 5.1 are more powerful than a visual comparison of these intervals would suggest.

Table A2. Percentile-bootstrap 95% confidence intervals (B=10,000, resampling reviews) for each system on the 609-example evaluation set. Estimates match Table 4.

System	ATE exact F1	ATE partial F1	Pair F1	Sent. Acc.
QLoRA fine-tuned LLMs				
Qwen 2.5-7B	0.708 [0.674, 0.740]	0.802 [0.775, 0.829]	0.645 [0.608, 0.680]	0.911 [0.886, 0.935]
Llama 3.1-8B	0.704 [0.670, 0.737]	0.801 [0.773, 0.828]	0.651 [0.615, 0.687]	0.925 [0.901, 0.947]
DeepSeek-R1-7B	0.678 [0.642, 0.714]	0.787 [0.758, 0.817]	0.603 [0.566, 0.640]	0.891 [0.860, 0.918]
Fine-tuned Uzbek encoders				
BERTbek	0.692 [0.657, 0.726]	0.800 [0.774, 0.826]	0.608 [0.571, 0.644]	0.879 [0.851, 0.906]
TahrirchiBERT	0.701 [0.667, 0.734]	0.787 [0.758, 0.814]	0.623 [0.586, 0.657]	0.889 [0.862, 0.915]

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